

# Jyotiranjana Beuria | Cover Letter

**Application for a Faculty Position at ICTS-TIFR, Bengaluru**  
jyotiranjana.beuria@gmail.com | 32/7, West Punjabi Bagh, New Delhi - 110026

Dear Members of the Faculty Search Committee,

I am writing to apply for a faculty position at the International Centre for Theoretical Sciences, Tata Institute of Fundamental Research, Bengaluru. My research lies at the interface of non-equilibrium statistical physics, active and biological matter, complex systems, and physics-inspired dynamical computation. I see a strong fit with the Biological Physics and Physics of Life community at ICTS, where living systems are studied through quantitative, physical, and computational principles.

My proposed research program, *Memory, Geometry, and Nonreciprocity in Adaptive Active Matter*, aims to develop a physics-grounded theory of adaptive non-equilibrium systems. By adaptive active matter, I mean many-body systems of self-driven units whose motility, interactions, and response rules are modulated by internal states, memory, feedback, and environmental context. The central goal is to understand how such internal-state dynamics generate synchronization, pattern formation, hysteresis, nonreciprocal response, and functional collective organization far from equilibrium.

The main contribution I can make to the *Physics of Life* direction is a theoretical and computational framework for biological collectives in which memory, feedback, nonreciprocity, and geometric constraints are treated as physical degrees of freedom rather than as phenomenological additions. These ingredients are central to living systems ranging from cells and tissues to animal groups and synthetic active agents. My work would aim to derive testable signatures such as delayed recovery after perturbation, path-dependent transitions, circulating probability currents, coherent modes, and topology-driven reorganizations of phase space.

My preparation for this program combines rigorous theoretical physics with a transition toward complex and living systems. During my Ph.D. at Harish-Chandra Research Institute under HBNI, I worked on high-energy phenomenology, vacuum structure, and collider physics, which trained me in model construction, numerical simulation, and high-dimensional data analysis. I later developed geometric and topological tools for physical data, including persistent homology and Forman-Ricci curvature in collider phase spaces, and then extended related ideas to biological aggregation, active matter, quantum-inspired collective motion, and nonreciprocal dynamics. My recent and ongoing works include publications in Physical Review D, Royal Society Open Science, Proceedings of the Royal Society A, and Physical Review E, along with under-review work on non-Markovian collective motion and context-sensitive dynamics.

At ICTS, I would develop a research program on memory-bearing active matter, context-driven nonreciprocal interactions, and geometric/topological diagnostics of high-dimensional collective dynamics. The work would focus on how internal states, environmental structure, and graph-based constraints reshape collective organization in active, soft, and biological systems, while also opening longer-term directions toward physics-constrained computation. This program would connect naturally with ICTS strengths in biological physics, soft matter, complex systems, and data-driven theoretical science.

I can also contribute to teaching and training at both core and interdisciplinary levels. I am prepared to teach statistical mechanics, classical mechanics, quantum mechanics, mathematical methods, computational physics, and nonlinear dynamics, and to develop advanced electives on active matter and living systems, geometric methods in physics, topological data analysis, and physics-inspired dynamical computation. My mentoring style emphasizes analytical clarity, reproducible computation, and movement between theory, simulation, and data interpretation.

I would be grateful for the opportunity to discuss how my research program in adaptive non-equilibrium physics and the physics of living systems could contribute to ICTS-TIFR. I enclose my curriculum vitae which includes publication list and a research statement for your consideration.

Sincerely,  
Jyotiranjana Beuria

# Memory, Geometry, and Nonreciprocity in Adaptive Active Matter

## From Active Matter to Physical Computing Architectures

### 1. Research Vision and Core Philosophy

My long-term research program aims to develop a physics-grounded theory of adaptive non-equilibrium systems. By *adaptive active matter*, I mean many-body systems of self-driven units whose motility, interactions, and response rules are dynamically modulated by internal states, memory, feedback, or environmental context. In this formulation, adaptation is not an external control layer imposed on an otherwise fixed active-matter model; rather, it is treated as a physical degree of freedom that co-evolves with collective motion and reshapes macroscopic organization.

The central question I address is how internal-state dynamics generate new forms of synchronization, pattern formation, nonreciprocity, hysteresis, and functional collective response far from equilibrium. This framework is motivated by biological and synthetic systems in which coherent organization emerges not only from local alignment or pairwise forces, but also from history-dependent response, context-sensitive interaction, and multi-timescale regulation.

Rather than using “collective intelligence” as a metaphor borrowed from artificial intelligence, my approach formulates adaptive collective behavior as a problem in non-equilibrium statistical mechanics, nonlinear dynamics, and geometric organization. The goal is to understand how memory-bearing, context-sensitive interacting units produce robust collective states, and how the same principles can eventually inform physics-constrained architectures for computation, including emerging directions in physical artificial intelligence.

To address this question, I combine tools from active matter, non-equilibrium statistical physics, discrete geometry, topological data analysis, dynamical systems theory and quantum theory. The resulting program builds a coherent pipeline from foundational physics: memory, nonreciprocity, geometry, and collective organization, to long-term applications in physical and neuromorphic computing.

### 2. Research Trajectory & Foundational Preparation

My theoretical foundation began in high-energy physics phenomenology and mathematical physics during my Ph.D. at Harish-Chandra Research Institute (HBNI). Working on supersymmetric phenomenology, vacuum structure, and high-dimensional computational event analysis resulted in six peer-reviewed publications (e.g., JHEP, CPC). This rigorous training equipped me with advanced skills in field-theoretic modeling, high-dimensional data processing, and numerical simulations.

There was a short break for around four years from formal academia in which I established my independent research center and collaboration for studies in consciousness and computational

neuroscience. This period led me to complex systems, computational neuroscience, and the physics of adaptive organization. Gradually, I pivoted toward geometry-driven non-equilibrium dynamics, systematically porting advanced mathematical tools to new domains:

- **Geometric & Topological Data Analysis (TDA):** In Physics Letters B (2023) and Physical Review D (2024), I have demonstrated the use of persistent homology and Forman-Ricci curvature to characterize high-dimensional collider phase spaces, demonstrating that discrete geometry can serve as an interpretable, intrinsic order parameter rather than a black-box analytical tool.
- **Active Matter & Nonreciprocity:** I have extended this geometric framework to biological and active systems. In a Royal Society Open Science (2024) paper, I have modeled biological aggregation driven by discrete Ricci curvature. Most notably, my recent paper in Physical Review E, "Topological Flux on a Context Manifold Generates Nonreciprocal Collective Dynamics," has established a concrete mathematical mechanism showing how nonreciprocal (asymmetric) forces emerge naturally when agents navigate a structured, underlying context manifold. I have also modelled active matter dynamics considering a quantum-inspired decision-making process (Proceedings of Royal Society A Journal, 2025).
- **Memory & Non-Markovian Systems:** In an under-review manuscript (arXiv:2510.23688), I have introduced a multi-timescale quantum-inspired framework where slow internal variables self-regulate fast spatial collective motion, revealing how history-dependent dynamics alter macroscopic synchronization and phase transitions. I foresee great potential applications in neuromorphic computing and drone swarming. I am working towards it.
- **Complex Biological Signals:** As a secondary testbed, I have co-authored multiple papers applying TDA, Hodge spectral entropy, and deep learning to noisy, non-stationary physiological signals (EEG/HRV) to decode cognitive state transitions and neural correlates of agency (BSPC 2024, IEEE Sensors 2026, PLOS ONE 2025).

### 3. Future Research Proposal

My research framework operates on the premise that real-world adaptive collectives rely on three fundamental ingredients typically omitted from minimal models: internal memory variables, nonreciprocal couplings, and geometry-dependent contextual constraints. I propose to systematically investigate these ingredients through four interconnected research aims.

#### Aim 1. Memory-Bearing Active Matter and Non-Markovian Collective Dynamics

Standard active-matter frameworks (e.g., Vicsek or Toner-Tu models) largely assume memoryless agents. However, biological and engineered units filter inputs, adapt response thresholds, and maintain internal states.

- **Objective:** I am exploring phase diagrams and continuum hydrodynamic limits for active matter where agents possess internal dynamical variables (e.g., perceptual uncertainty, attention-like gain, or regulatory tone).

- Expected Outcomes: Quantifying how non-Markovian memory shifts universality classes, introduces hysteresis, creates path-dependent order-disorder transitions, and generates anomalous relaxation kinetics under external perturbations.

## **Aim 2. Nonreciprocal Dynamics Emergent from Contextual Interaction**

Nonreciprocity breaks detailed balance and can generate circulating probability currents, traveling modes, chiral patterns, and functional steady states that have no equilibrium analogue. In many biological, cognitive, and engineered systems, however, nonreciprocal coupling is not simply imposed externally; it can emerge because interacting units respond to a shared context differently.

Objective: Building on my framework published in *Physical Review E*, I will develop minimal agent-based and hydrodynamic models in which directed, asymmetric interactions arise from contextual modulation rather than from ad hoc nonreciprocal rules. “Context” is the structured background of internal states, environmental conditions, relational constraints, and prior history that determines how a system interprets inputs and generates effective interactions. The goal is to understand how such context-dependent interaction rules generate effective nonreciprocity, and how this nonreciprocity reorganizes collective motion, synchronization, pattern formation, and stability.

This aim will also connect with my broader work on context-sensitive dynamics, including an under-review study on emergent non-classical probabilistic structure in LLMs under contextual modulation. While that work belongs to a different substrate, it motivates a common theoretical question: how does context reshape the effective state space, transition structure, and collective response of an adaptive system? Subsequent studies will explore this question across synthetic active agents, computational systems, and living collectives.

Expected Outcomes: This aim will produce a classification of context-driven nonreciprocal dynamics, identifying when contextual asymmetry stabilizes coherent functional organization and when it produces oscillations, traveling instabilities, frustration, or chaotic behavior. Particular attention will be given to systems in which nonreciprocity interacts with multi-timescale memory, since this coupling is expected to generate path-dependent transitions, hysteresis, and adaptive reconfiguration.

## **Aim 3. Geometric and Topological Diagnostics of High-Dimensional Phase Spaces**

Complex non-equilibrium systems often undergo structural re-organization that eludes conventional scalar order parameters like mean polarization or density.

- Objective: I am building a robust library of geometric and topological order parameters, leveraging persistent homology, Forman- and Ollivier-Ricci curvatures, graph Laplacians, and Hodge decomposition, tailored for active matter and complex network trajectories.
- Expected Outcomes: Developing open-source diagnostic pipelines capable of identifying phase-space boundary crossings, critical slowing down, coherent structures, and latent

topological defects in both simulated models and experimental soft-matter or biological datasets.

**Aim 4. From Non-Equilibrium Physics to Physics-Constrained Physical Computing**

The ultimate phase of this program bridges the physics of self-organization with the engineering of alternative computational substrates, focusing on reservoir computing (RC).

- Objective: I am designing unconventional, active-matter-inspired reservoirs characterized by intrinsic fast/slow variables in quantum formalisms, nonreciprocal graph topologies, and explicit conservation/boundedness constraints. These systems will be benchmarked against standard recurrent architectures (LSTMs, ESNs) on low-data temporal tasks and rapid context-switching environments. I have an ongoing manuscript under review in an IEEE Transactions venue in this direction.
- Long-Term Vision: Partnering with experimental labs to realize these design principles in physical substrates, such as neuromorphic, electronic, optical, or soft active media.

**4. Deliverables**

|         | Research Focus              | Key Milestones & Deliverables                                                                                                                                           |
|---------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 | Foundational Physics        | Phase diagrams of memory-bearing active matter; continuum limits; analytical models of circulating probability flows.                                                   |
| Phase 2 | Geometric Diagnostics       | Deployment of TDA and graph-curvature software pipelines to analyze non-equilibrium phase-space boundaries, particularly in real-life scenarios such as drone swarming. |
| Phase 3 | Computational Architectures | Design and benchmarking of multi-timescale nonreciprocal reservoirs on temporal and context-switching tasks.                                                            |
| Phase 4 | Consolidation & Realization | Translating principles into physical/neuromorphic hardware substrates via collaborative experimental testbeds.                                                          |

**5. Collaboration, Mentorship, and Institutional Integration**

While my core approach is theoretical and computational, it is designed to act as a collaborative hub:

- **Experimental Integration:** The topological and geometric diagnostic tools developed in Aim 3 are directly applicable to tracking data from soft-matter physics, synthetic active agents, and biological networks.
- **Teaching & Training:** I am fully prepared to teach core curricula (Statistical Mechanics, Classical Dynamics, Mathematical Physics, Quantum Mechanics, Quantum Field Theory, etc.) alongside specialized electives such as The Physics of Complex Systems, Topological Methods in Physics, and Physics-Inspired Dynamical Computation. Student mentorship in my group will emphasize analytical rigor balanced with open-source scientific computing.

## 6. Conclusion

Over the next decade, my goal is to transform the concept of adaptive collective intelligence from a descriptive phenomenon into a predictive, exact science. By demonstrating how memory, geometry, and nonreciprocity dictate the boundaries of physical self-organization, this research program will lay the groundwork for a generation of computing systems whose intelligence arises naturally from physical law rather than unconstrained parameter optimization.

### Applicant's representative publications (full list is in CV):

- **J Beuria**, Emergent Non-Classical Probabilistic Structure in Large Language Models Under Contextual Modulations. 2026. <https://ssrn.com/abstract=6504379> (Under review)
- **J Beuria**, and V H Chembrolu. "Topological Flux on a Context Manifold Generates Nonreciprocal Collective Dynamics." **Physical Review E** 113, no. 6 (June 2026): 065401.
- **J Beuria**, et al. Collective motion from quantum-inspired dynamics in visual perception. **Proceedings of the Royal Society A Journal: Mathematical, Physical and Engineering Sciences**. 2025;481(2321):20250489. arXiv:2409.18985.
- **J Beuria**, L Behera. Non-local interaction in discrete Ricci curvature-induced biological aggregation. **Royal Society Open Science**. 2024;11(9):240794.
- **J Beuria**. Non-Markovian collective motion from self-regulated perceptual dynamics. arXiv:2510.23688. **Under review**.
- **J Beuria**. "Probing Invisible Decay of Z' at Muon Collider with Topological Data Analysis and Machine Learning." arXiv:2509.19756 (**under review**).
- **J Beuria**. "Topological Landscapes of the BSM Higgs Sector." arXiv:2510.10900 (**under review**).
- **J Beuria** (2023). "Persistent homology of collider observations: When (w) hole matters." *Physics Letters B*, 846, 138188.
- **J Beuria (2024)**. "Intrinsic geometry of collider observations and Forman Ricci curvature." *Physical Review D*, 110(3), 035023.
- K V Gupta, **J Beuria**, L Behera. Characterizing EEG signals of meditative states using persistent homology and Hodge spectral entropy. *Biomedical Signal Processing and Control*. 2024;89:105779.
- S Singh, **J Beuria**, L Behera, Ram Bilas Pachori, Vinay Gupta. Decoding cognitive load changes induced by mantra meditation from physiological signals using deep neural networks. *IEEE Sensors Journal*. 2026;26(1):1088–1102. doi:10.1109/JSEN.2025.3628683.
- K V Gupta, **J Beuria**, L K Vijanapalli, A Sethi, L Behera. Self-reflection, sense of agency, and underlying neural correlates: A pilot study. *PLOS ONE*. 2025;20(12):e0335276. doi:10.1371/journal.pone.0335276.



# JYOTIRANJAN BEURIA

DOB: 15<sup>th</sup> May 1990 | jyotiranjana.beuria@gmail.com | 32/7, West Punjabi Bagh, New Delhi-26

## Research Interest

- Physics of Living Systems & Complex Systems
- Machine Learning and Applications
- Computational Cognitive Neuroscience
- High Energy Physics (Phenomenology)

## Education

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|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>PhD (High Energy Physics)</b><br>Homi-Bhabha National Institute<br>(Harish-Chandra Research Institute, Prayagraj)<br>(Awarded Infosys Scholarship-2017) | 2012-2018 |
| <b>Integrated M.Sc Physics</b><br>Indian Institute of Technology, Roorkee<br>(Awarded Institute Silver Medal-2012)                                         | 2007-2012 |
| <b>+2 Science</b><br>Ravenshaw Junior College, Cuttack                                                                                                     | 2005-2007 |
| <b>Secondary</b><br>Saraswati Vidya Mandir, Salipur, Cuttack<br>(Rank-3 in Secondary Board of Odisha)                                                      | 2005      |

## Work Experience

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Chief Scientist</b><br>manasai.tech                                                                                                                        | Mar 2026 - Present  |
| <b>EEG and Cognitive Neuroscience SME</b><br>Brainwave Science Pvt. Ltd.                                                                                      | Sep 2025 - Feb 2026 |
| <b>Post Doctoral Fellow</b><br>Indian Institute of Technology, Mandi, Himachal Pradesh                                                                        | July 2022-July 2025 |
| <b>Principal Investigator</b><br>IKS Research Center, ISS Delhi<br>(Recognised as IKS Research Center by IKS Division, Ministry of Education, Govt. of India) | Jan 2018-present    |

## Journals

### Physics of Living Systems & Complex Systems

1. **Jyotiranjana Beuria**, Emergent Non-Classical Probabilistic Structure in Large Language Models Under Contextual Modulations. 2026. <https://ssrn.com/abstract=6504379> (Under review)
2. **Jyotiranjana Beuria**, and Venkatesh H. Chembrolu. "Topological Flux on a Context Manifold Generates Nonreciprocal Collective Dynamics." **Physical Review E** 113, no. 6 (June 2026): 065401 (Q2).
3. **Jyotiranjana Beuria**, Mayank Chaurasiya, Laxmidhar Behera. Collective motion from quantum-inspired dynamics in visual perception. **Proceedings of the Royal Society A Journal: Mathematical, Physical and Engineering Sciences**. 2025;481(2321):20250489. arXiv:2409.18985. (Q1)
4. **Jyotiranjana Beuria**, Laxmidhar Behera. Non-local interaction in discrete Ricci curvature-induced biological aggregation. **Royal Society Open Science**. 2024;11(9):240794. (Q1)
5. **Jyotiranjana Beuria**. Non-Markovian collective motion from self-regulated perceptual dynamics. arXiv:2510.23688. Under review.
6. **Jyotiranjana Beuria**, Venkatesh H. Chembrolu, Nitin Kumar. A Formal Framework for Phenomenal Unity, Contextual Experience, and Intersubjective Coherence. 2026. <https://philpapers.org/rec/BEUACF>, Under review.

## High Energy Physics

1. **Jyotiranjana Beuria**. "Probing Invisible Decay of  $Z'$  at Muon Collider with Topological Data Analysis and Machine Learning." arXiv:2509.19756 (**under review**).
2. **Jyotiranjana Beuria**. "Topological Landscapes of the BSM Higgs Sector." arXiv:2510.10900 (**under review**).
3. **Jyotiranjana Beuria** (2023). "Persistent homology of collider observations: When (w) hole matters." Physics Letters B, 846, 138188. (**Q1**)
4. **Jyotiranjana Beuria** (2024). "Intrinsic geometry of collider observations and Forman Ricci curvature." Physical Review D, 110(3), 035023. (**Q1**)
5. **Jyotiranjana Beuria**, AreshKrishna Datta, Dipsikha Debnath, and Konstantin T. Matchev (2018). "LHC collider phenomenology of minimal universal extra dimensions." Computer Physics Communications, 226, 187–205. (**Q1**)
6. **Jyotiranjana Beuria** and Abhishek Dey (2017). "Exploring charge and color breaking vacuum in non-holomorphic MSSM." Journal of High Energy Physics, 2017(10), 1–27. (**Q1**)
7. **Jyotiranjana Beuria** and AreshKrishna Datta (2017). "Spontaneous breakdown of charge in the MSSM and in the NMSSM: Possibilities and implications." Journal of High Energy Physics, 2017(11), 1–26. (**Q1**)
8. **Jyotiranjana Beuria**, Utpal Chattopadhyay, AreshKrishna Datta, and Abhishek Dey (2017). "Exploring viable vacua of the  $Z_3$ -symmetric NMSSM." Journal of High Energy Physics, 2017(4), 1–52. (**Q1**)
9. **Jyotiranjana Beuria**, Arindam Chatterjee, and AreshKrishna Datta (2016). "Sbottoms of natural NMSSM at the LHC." Journal of High Energy Physics, 2016(8), 1–31. (**Q1**)
10. **Jyotiranjana Beuria**, Arindam Chatterjee, AreshKrishna Datta, and Santosh Kumar Rai (2015). "Two light stops in the NMSSM and the LHC." Journal of High Energy Physics, 2015(9), 1–47. (**Q1**)

## Neuroimaging and Application of AI

1. Kurusetti Vinay Gupta, **Jyotiranjana Beuria**, Laxmidhar Behera. Characterizing EEG signals of meditative states using persistent homology and Hodge spectral entropy. Biomedical Signal Processing and Control. 2024;89:105779. (**Q1**)
2. Swati Singh, **Jyotiranjana Beuria**, Laxmidhar Behera, Ram Bilas Pachori, Vinay Gupta. Decoding cognitive load changes induced by mantra meditation from physiological signals using deep neural networks. IEEE Sensors Journal. 2026;26(1):1088–1102. doi:10.1109/JSEN.2025.3628683. (**Q1**)
3. Kurusetti Vinay Gupta, **Jyotiranjana Beuria**, Lokeshwara Kumar Vijanapalli, Amit Sethi, Laxmidhar Behera. Self-reflection, sense of agency, and underlying neural correlates: A pilot study. PLOS ONE. 2025;20(12):e0335276. doi:10.1371/journal.pone.0335276. (**Q1**)
4. Durgesh Ameta, Ankit Yadav, **Jyotiranjana Beuria**, Laxmidhar Behera. Multimodal odor perception prediction using olfactory EEG and physicochemical features of odorants. ChemRxiv. 2025. doi:10.26434/chemrxiv-2025-fbmlf. Preprint; *Manuscript under review*.
5. Swati Singh, **Jyotiranjana Beuria**, Ram Bilas Pachori, & Laxmidhar Behera. Neural and affective markers of kirtan meditation: Persistent homology of multivariate EEG in real and immersive environments. 2026. *Manuscript under review*.
6. **Jyotiranjana Beuria**. Topological and graph-theoretic analysis of EEG dynamics in mind-wandering and focused attention. 2025. *Manuscript under review*.



## Book Chapters

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Kurusetti Vinay Gupta, **Jyotiranjana Beuria**, and Laxmidhar Behera. "Chapter 14 — Neuroscience of Meditation for Holistic Well-Being." Biological Measures of Well-Being, edited by Colin R. Martin, Vinood B. Patel, Rajkumar Rajendram, and Victor R. Preedy, Academic Press, 2026, pp. 175–188. ScienceDirect, <https://doi.org/10.1016/B978-0-443-28842-5.00001-6>.

## Organisation of Conferences

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- Organised a one-day conference on **"Unriddling Inference: From Pramāṇa Theory to Modern Logic and AI"** at IIT Delhi. 12 April 2026
- Organised a one-day conference on **"The Enigma of Perception: Perspectives from Indian Knowledge System"** at IIT Delhi. 24 August 2025
- Part of the organising team of the conference **"Mind, Brain and Consciousness Conference (MBCC)-2025"**, held at IKSMHA Centre, IIT Mandi. 4-7 June 2025
- Part of the organising team of the conference **"Mind, Brain and Consciousness Conference (MBCC)-2023"** held at IKSMHA Centre, IIT Mandi. 14-16 December 2023
- Organised **"Mind, Matter, and Consciousness: From Information to Meaning"**, International Virtual Conference. 18-19 Dec 2021

## Grants

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- IKS Research Center grant for 2023-2025 **(35.27 lakh INR ~ 37,400 USD )**  
(IKS Division, Ministry of Education, Govt. of India)
- IKS Research Center grant for 2025-2027 **(10 lakh INR ~ 10,600 USD )**  
(IKS Division, Ministry of Education, Govt. of India)

## Teaching Experience

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- Introduction to Topological Data Analysis
- Neuroimaging Techniques and Signal Processing
- Introduction to Quantum Mechanics
- Physics of Collective Motion

## Other Achievements

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- **Winner (IKS Category)** in the Himalayan Start-up Trek held at IIT Mandi, Jan 2026
- **Second Prize at Global Indology Conclave 2026** for the research-based documentary "Perception and Reality in Indian Knowledge System".
- **Infosys Scholarship** in High Energy Physics-2017, HRI Allahabad
- Graduate Aptitude Test in Engineering (GATE)-2012, All India Rank-13
- Joint Entrance Screening Test (JEST)-2012, All India Rank-20
- Council of Scientific and Industrial Research (CSIR)-National Eligibility Test (NET)- Junior Research Fellowship (JRF)-2011, All India Rank-38
- **Institute Silver Medal-2012**, Indian Institute of Technology, Roorkee
- Working Internships in Science and Engineering (**WISE**)-**2010 Scholarship**, Deutscher Akademischer Austauschdienst, Germany
- IIT-JEE 2007 AIR: 3894 (**98.5 percentile**) and AIEEE 2007, AIR: 2008 (**99.6 percentile**)
- Orissa Governor's Award for holding **3rd rank** in Odisha Board in High School Certificate Examination-2005